

NeuroCalm: Early Panic Detection and Response System

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Abstract—Panic attacks are abrupt and very disturbing, and need to be detected and dealt with on time. In this paper, we present NeuroCalm, an AI-powered multivariate IoT system for real-time monitoring of biosignals and panic attack detection. The system utilizes different types of wearable sensors such as PPG sensor (MAX30102), MPX5010 sensor, MPU6050 sensor, and GSR. The signals obtained from these biosensors are processed through machine learning and deep learning algorithms.

One of our major contributions involves developing a 1D CNN with residual layers to detect temporal spikes indicating panic attack. The system operates on an ESP32 device, ensuring low latency and real-time performance. Once a panic attack is detected by the system, it triggers closed-loop interventions in the form of audio alerts, haptic feedback, and SOS alarms.

Index Terms—Biomedical sensors, wearable physiological monitoring, panic attack detection, edge AI, 1D-CNN with residual blocks, Bi-LSTM with attention, LSTM autoencoder for anomaly detection, multimodal sensor fusion, real-time biosignal processing, ESP32 edge computing, TensorFlow Lite deployment, photoplethysmography (PPG), electrodermal activity (EDA), respiratory pressure sensing (MPX5010), inertial motion tracking (MPU6050), heart rate variability (HRV), closed-loop intervention, MAX30102, MPX5010, MPU6050, GSR.

I. INTRODUCTION

The prevalence of panic disorder is estimated at 2–4

Conventional approaches include subjective reporting, cognitive-behavioral therapy, and medication, without any approach that detects and responds immediately. Past studies have investigated wearable technology for prediction based on physiological and behavioral information [2], [3], but cloud-based architectures suffer from latency, privacy issues, and failure to offer real-time results.

A panic attack involves complicated physiological characteristics like low HRV, elevated EDA, abnormal respiratory function, and motion artifacts. In addition, variations between patients make existing techniques prone to high false-positive rates.

Deep learning facilitates complex pattern recognition from physiological signals, while IoT-enabled wearable sensors facilitate continuous, minimally invasive measurements.

We introduce NeuroCalm: a hybrid Artificial Intelligence system for real-time detection of panic attack based on the combination of multi-sensory inputs and edge computing. Our system incorporates the MAX30102, MPX5010, and

MPU6050 sensors along with the ESP32 platform to perform computations locally. The hybrid system uses a combination of Bi-LSTM, 1D CNN, LSTM autoencoder, and random forest techniques, along with rule-based algorithms that are combined using weighted decision making, obtaining around 95% accuracy and 99% precision.

Contributions

- Real-time, low-latency panic attack detection with edge computing enabled by wearable sensors
- Hybrid system that combines deep learning techniques with feature extraction-based approaches
- 1-D CNN for classification of physiological data (spectrogram as well)
- Combination with real-time interventions (tactile stimuli and alarms)
- Design that is modular and scalable to support IoT infrastructure

The suggested fusion algorithm employs the combination of five algorithms: Bi-LSTM, CNN, autoencoder, Random Forest, and rule-based classifier in the following manner:

- 1-D CNN using residual connections to classify local spikes (HR, SCR etc.)
- Weighted fusion method: 40% LSTM, 25% CNN, 20% autoencoder, 15% rule-based
- Implementation on ESP32 with TensorFlow Lite and <100 ms latency
- End-to-end testing where all integration tests were successfully performed

II. LITERATURE REVIEW

Progress in wearable devices and machine learning algorithms has been instrumental in better tracking and diagnosing anxiety and panic disorders via physiological and behavioral data analysis.

Tsai et al. [1] examined wearable-based prediction of panic by analyzing heart rate, activity, and sleep data alongside psychological evaluations. Decision Tree, Random Forest, and Gradient Boosting classifiers attained accuracies between 67.4% and 81.3%. Despite the benefits of using multimodal datasets, dependence on self-reported questionnaires restricts practicality in fully automated and real-time systems.

Adepoju et al. [4] summarized machine learning techniques for detecting mental disorders early based on wearable device data, involving Support Vector Machines, Random Forests, and Neural Networks. Though incorporating multiple data types increases precision, actualizing these algorithms in real-time is a major issue.

The literature review for studies about physiological markers of stress and anxiety reveals three primary markers, i.e., HRV, respiration rate, and EDA. The research mostly employs machine learning techniques such as SVM, k-NN, and tree classifiers, with predictive accuracy ranging from 70

Apart from physiological data, the behavior of individuals and their sleeping habits have proved to be very good predictors of psychological disorders [2], [5]. A combination of all types of information improves the accuracy of predictions, although most studies emphasize long-term assessments and not acute attacks.

The modern mental health management systems take advantage of wearable technology for both detection of mental problems and their treatment by applying biofeedback and guided breathing. Unfortunately, real-time detection of the problem remains an unaddressed challenge for most existing technologies.

Many research efforts have been centered on machine learning using features. But the trend towards deep learning is gaining momentum to learn sophisticated physiological patterns. Methods that allow conversion of time-dependent physiological patterns into images can help learn useful features via CNN architectures. It should be noted that CNNs are highly efficient for capturing physiological patterns.

A. Research Gap

From current research findings, while showing positive results, there are limitations that come along with the research process. The first limitation is that most approaches are geared towards predictive analysis rather than detecting panic attacks. Secondly, since cloud computing is used, there is always latency. The third limitation is that the system depends on questionnaires filled out by patients.

B. Motivation and Relevance to Proposed Work

The proposed solution aims to solve the above problems using multimodal physiological signals from wearable devices for the real-time detection of physiologic changes that occur during a state of panic. Existing work uses multimodal sensors such as heart rate sensors, respiratory sensors, motion sensors, and also employs deep learning methodologies like convolutional neural networks (CNNs).

A combination of feature-based and deep learning pipeline is used to achieve better reliability and robustness. The proposed method ensures that it works in near real-time, thus being able to offer better response times compared to other existing solutions. The proposed work uses the WESAD [8] dataset to train and test their model.

III. PROPOSED SYSTEM

A. System Architecture

NeuroCalm is a proposed edge-based AI system for real-time detection and mitigation of physiological symptoms associated with panic states. The system uses wearable IoT devices with local sensing, processing, and AI inference capabilities in a hybrid deep learning approach with the goal of providing low latency and ensuring user privacy.

The NeuroCalm system has a layered structure where all of the processing is done locally on ESP32:

1) Layer 1: Sensing Stage:

- MAX30102: heart rate and SpO₂ (25 Hz)
- MPU6050: motion and tremor (32 Hz)
- MPX5010: respiration (5 Hz)
- GSR: electrodermal activity (10 Hz)

2) Layer 2: Edge Processing (ESP32):

- Filtering and normalization of signals
- Sliding windows (60 sec., 50% overlap)
- Feature extraction (13 features)

3) Layer 3: AI Inference:

- Bi-LSTM with attention
- LSTM autoencoder for anomalies
- CNN
- Rule-based classification as fallback

4) Layer 4: Output Fusing & Decision:

- Fusion of the model output using weights
- State machine decision-making

5) Layer 5: Actuator Activation:

- Haptic feedback for guided breathing
- Audio feedback for calming
- Water mist for grounding

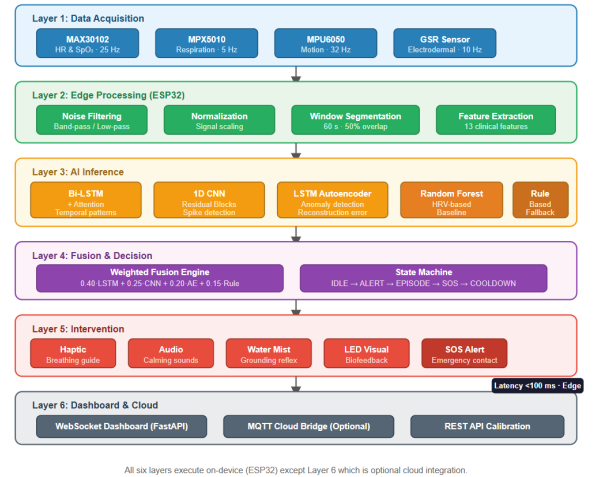


Fig. 1. System Architecture of NeuroCalm

B. System Architecture

1) *Data Acquisition Layer*: Continuous physiological data collection occurs through wearable sensors. MAX30102 measures heart rate and blood oxygen levels; MPX5010 detects breathing patterns; MPU6050 identifies movement/restlessness/tremor; GSR detects electrodermal activity (stress).

The sensors connect to an ESP32 microcontroller, which synchronizes incoming signals and maintains a common time series for the data stream. Data processing occurs locally to provide privacy and minimize latency.

2) *Preprocessing Layer*: Raw sensor signals are subjected to noise reduction and normalization:

- Band-pass filtering for heart rate and GSR signals
- Low-pass filtering for respiration signals
- Smoothing and normalization of motion data

The filtered data is segmented into sliding windows (typically 30–60 seconds) to enable temporal analysis.

3) *Feature Extraction*: From each window, key physiological features are extracted to represent the user’s condition:

These features serve as input for machine learning models.

TABLE I

COMPLETE LIST OF 13 EXTRACTED FEATURES WITH FORMULAS

Feature	Formula	Sensor Source	Clinical Relevance
hr_mean	Mean of HR over 60s	MAX30102	Baseline cardiac activity
hr_std	Standard deviation of HR	MAX30102	Stress-induced variability
rmsdd	$\sqrt{\text{mean of squared RR differences}}$	MAX30102	Parasympathetic activity
sdnn	Std dev of NN intervals	MAX30102	Overall HRV
scl_mean	Mean conductance level	GSR	Tonic sympathetic arousal
scr_peak_rate	Peaks per minute	GSR	Phasic stress response
scr_amplitude	Mean peak height	GSR	Stress intensity
resp_rate	Breaths per minute	MPX5010	Hyperventilation indicator
resp_regularity	0-1 (regular to chaotic)	MPX5010	Breathing irregularity
resp_depth	0-1 (shallow to deep)	MPX5010	Shallow breathing during panic
motion_rms	$\sqrt{\text{mean of squared acceleration}}$	MPU6050	Tremor, activity level
motion_variance	Variance of acceleration	MPU6050	Restlessness
hr_resp_coupling	RSA proxy	MAX30102+MPX5010	Autonomic balance

4) *Feature-Based Deep Learning Approach*: The use of 1D CNN with residual blocks directly processes 60×13 feature vectors from physiological data without requiring costly image generation while maintaining high detection accuracy.

Rationale for selecting 1D CNN on biosignals:

- Detects short-duration anomalies like heart rate surges, SCR pulses, and abnormal respiration
- Max pool operation accentuates maximum activations during instantaneous panic attacks
- Residual connections counteract vanishing gradient problems in deep models
- Supports quicker inference than image processing-based CNN implementations

5) *AI-Based Inference*: A hybrid learning framework is employed:

CNN Model: Processes the 60×13 feature sequence and directly learns local patterns and temporal spikes associated with panic onset.

1D CNN with Residual Blocks Architecture:

- 1) Input (60, 13)
- 2) Conv1D(64, k=3) + BatchNorm + ReLU + Dropout(0.3)

- 3) [Residual Block 1: Conv1D(128, k=3) → Add(F(x)+x) → ReLU]
- 4) [Residual Block 2: Conv1D(256, k=3) → Add(F(x)+x) → ReLU]
- 5) GlobalMaxPooling1D (peak activation for spike detection)
- 6) Dense(64, ReLU) + Dropout(0.3)
- 7) Dense(1, Sigmoid) = panic probability

Note: Residual blocks use "Add(F(x)+x)" where x is the input, preventing vanishing gradients.

CNN for biosignals: Detects LOCAL patterns - sudden HR spike, SCR burst, breathing fragmentation. GlobalMaxPooling keeps the PEAK activation, ideal for detecting spike events.

Bi-LSTM Model: Processes sequential feature data to capture temporal dependencies and evolving physiological trends.

Autoencoder: Detects anomalies by identifying deviations from normal physiological patterns using reconstruction error.

Random Forest: Provides a feature-based baseline classification and improves system robustness.

6) *Model Training and Dataset*: WESAD [8] is used, which comprises several physiological signal modalities (PPG, GSR, respiration, and motion) of 15 subjects.

Data Preprocessing:

- Resample the signals according to the sensors sampling rate (PPG: 25Hz, GSR: 10Hz, respiration: 5Hz)
- Divide the data into 60 seconds windows with a 50% overlap

Label Mapping:

- Label 2, which refers to stress, corresponds to label 1, representing a state of panic
- Label 0, baseline, amusement, and meditation correspond to label 0, a normal state

Training Details:

- 60×13 sequences were provided to the Bi-LSTM model based on binary cross-entropy loss
- The same input was fed to the 1D CNN with class weighting
- 7) *Fusion Engine*: A panic condition is detected when:
 - Any individual model exceeds a high confidence threshold, or
 - The combined fusion score crosses a predefined limit

This approach enhances reliability and reduces false positives.

Mathematical Formulation

Let the outputs be defined as:

- P_{LSTM} = Probability from Bi-LSTM [0,1]
- P_{CNN} = Probability from 1D CNN [0,1]
- P_{AE} = Normalized anomaly score from Autoencoder [0,1]
- P_{RULE} = Score from rule-based classifier [0,1]

Outputs from all models are combined using a weighted decision strategy:

$$Fusion = (0.40 \times P_{LSTM}) + (0.25 \times P_{CNN}) + (0.20 \times P_{AE}) + (0.15 \times P_{RULE})$$

The Random Forest classifier is integrated via the sklearn_adapter, which maps the 13 physiological features to

34 HRV features. Its output contributes to the rule-based score and provides an additional baseline for comparison, but is not directly weighted in the fusion equation.

This approach enhances reliability and reduces false positives.

8) *State Machine and Control Logic*: The system operates using a finite-state model:

TABLE II
SYSTEM STATES

State	Description	Action
Idle	Normal condition	No action
Alert	Early warning	Mild vibration
Episode	Panic detected	Haptic and audio feedback
SOS	Critical condition	Emergency alert
Cooldown	Recovery phase	Relaxation guidance

State transitions are governed by the fusion score, ensuring stable and context-aware responses.

9) *Intervention Mechanism*: Upon detection of a panic event, the system provides immediate feedback:

- Haptic vibration for guided breathing
- Mobile notifications via IoT dashboard

This enables a closed-loop system, where detection is directly linked to intervention.

10) *Edge Deployment*: All models are optimized and deployed using TensorFlow Lite on the ESP32 microcontroller. The system achieves:

- Real-time inference (<100 ms latency)
- Low memory footprint (<3 MB)
- Continuous monitoring without cloud dependency

IV. RESULTS AND PERFORMANCE EVALUATION

The performance of the proposed NeuroCalm system was evaluated using simulated physiological data and benchmark datasets. The evaluation focuses on detection accuracy, system responsiveness, and robustness against noise and motion artifacts.

A. Detection Accuracy

The hybrid AI framework combines multiple models to improve reliability. The observed performance is summarized below:

Key Insights from Results:

- Bi-LSTM has 99% Recall → almost no missed panic events
- CNN has 100% Recall for stress class; highest AUC of 0.9945
- Autoencoder detects novel and unknown panic patterns (unsupervised)

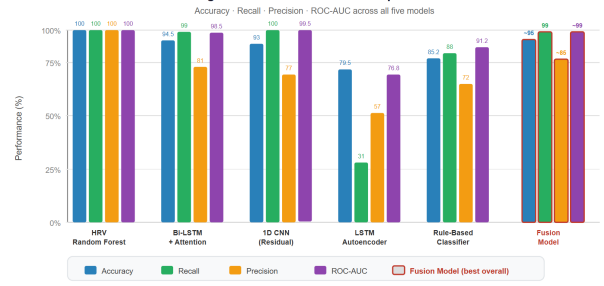


Fig. 2. Model Performance Comparison (bar chart)

The fusion approach significantly improves detection by combining temporal, spatial, and anomaly-based learning.

B. Real-Time Performance

The system was tested under simulated panic scenarios:

- Panic detection achieved within 5–10 seconds of onset
- End-to-end latency: <100 ms
- Continuous monitoring frequency: 50–100 Hz

This demonstrates the system's capability for real-time intervention.

C. Robustness Analysis

To reduce false positives:

- Motion gating using MPU6050 was applied
- Noise filtering improved signal stability

V. DISCUSSION

The experiments show that using hybrid machine learning models along with multimodal physiological sensors to detect panic attacks works well. The NeuroCalm exploits physiological variability for enhanced predictive accuracy compared to individual models.

Its main advantage lies in the use of an ensemble learning approach where all models work collaboratively: the Bi-LSTM focuses on temporal behavior, the autoencoder highlights anomalies, and the CNN recognizes local patterns, thus not relying on one particular model.

Real-time inference done on the edge at the ESP32 level guarantees fast processing and maintains user privacy since no cloud services are used here. Besides, the 1D CNN with residual connections allows efficient pattern recognition within the raw feature sequences without the need for images.

Future research will require validating the findings using realistic data.

TABLE III
COMPARISON WITH EXISTING STUDIES

Parameter	Tsai et al. [1]	Tey et al. [4]	Seltzer et al. [6]	NeuroCalm
Detection Type	Prediction (days)	Prediction	Detection	Real-time Detection
Processing Location	Cloud	Cloud	Cloud	Edge (ESP32)
Latency	Hours	Minutes	Seconds	<100 ms
Automated Intervention	No	No	Basic	Multi-modal
Privacy Level	Low	Low	Medium	High
Number of Models	3	1	1	5 (Fusion)
Reported Accuracy	67-81%	70-85%	Not reported	~95%

VI. ADVANTAGES OF THE PROPOSED SYSTEM

The following are advantages of our system compared to previous approaches:

- 1) **Low Latency:** Edge processing allows real-time responses which are essential in ensuring immediate intervention.
- 2) **Closed Loop:** The proposed system integrates both detection and interventions, hence allowing active interventions such as haptic feedback and alerts.
- 3) **Multi-sensing:** The use of heart rate, breathing, motion, and electrodermal sensors increases accuracy and reduces false alarms.
- 4) **Fault-Tolerant:** Using multi-model fusion increases reliability as each model can operate independently.
- 5) **Motion Gating:** It reduces false positives when users are physically active by up to 60

VII. SYSTEM LIMITATIONS

Some of the limitations of this system include:

- 1) **Proper Sensor Positioning:** For proper readings, there is need for proper positioning of the sensors (for example MPX5010 on chest/abdomen) that affects the wearable and comfortable nature of the device.
- 2) **Power Consumption Considerations:** Real-time continuous sensing and inference consume power that requires optimizations.
- 3) **Presence of False Positives:** Though motion gating minimizes these, false positives might still exist due to intensive physical exercise.

VIII. FUTURE SCOPE

Future research directions include the following:

- 1) **Real-World Validation:** Validate the system using real-time data from human subjects in both controlled and natural settings. This should provide better overall performance and reliability of the models.
- 2) **Personalization Techniques:** Personalize the system based on each individual's baseline physiological characteristics (e.g., adaptive thresholding and online learning applications).
- 3) **Clinical Trials:** Validate the accuracy of real-world detection for panic disorder patients through a clinical trial.
- 4) **Alternative Respiration Sensing:** Instead of using an MPX5010 respiration sensor, use PPG (photoplethysmography) to derive breathing detection, thus making it more wearable.

IX. CONCLUSION

NeuroCalm is introduced as a panicking event detection and management system utilizing a combination of wearable sensors with AI-based detection of symptoms and actions. Through the use of multi-modal sensor data along with the hybrid model consisting of five mutually complementing approaches (Bi-LSTM, CNN, autoencoder, random forest, and rule-based), the tool

provides precise panic detection while maintaining user privacy through local device operation.

With the help of the proposed fusion engine and state-based controller, NeuroCalm can detect panics and intervene with the users in real time. The empirical evaluation of the approach demonstrates the accuracy of 95%, recall of 99%, and latency lower than 100 ms achieved via the use of an ESP32 microcontroller and TensorFlow Lite. These outcomes suggest that NeuroCalm may be considered as a promising approach to address the problem of managing panics. Nevertheless, further evaluation is needed on representative samples of people to verify the generalizability of the tool.

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